Stat 286: Causal Inference with Applications, Fall 2025

Section 3: Average Treatment Effects

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1 Overview

We distinguish two complementary perspectives for average treatment effects (ATEs): the finite-population (design-based) view and the super-population (model/sampling-based) view.

Finite-population (design-based) view. We fix the n units in our study. By unit-level determinism (SUTVA/consistency assumed), each unit i has fixed potential outcomes $\{Y_i(0), Y_i(1)\}$; the only randomness comes from the assignment mechanism. The causal estimand of interest is the *sample* ATE (SATE),

$$\tau_{\text{SATE}} \;=\; \frac{1}{n}\sum_{i=1}^{n}\big(Y_i(1) - Y_i(0)\big).$$

Super-population (model-based) view. We imagine the units are an i.i.d. sample from a large (possibly infinite) population where $(Y(1), Y(0)) \sim \mathbb{P}^*$ are random. Randomness comes from the treatment assignment and the sampling. The causal estimand of interest is the *population* ATE (PATE)

$$\tau_{\text{PATE}} \;=\; \mathbb{E}[Y(1) - Y(0)]\,.$$

2 Comparison

3 Results

Lemma 1 (Unbiasedness of the difference-in-means estimator) *Let* $\hat{\tau} = \overline{Y}_1 - \overline{Y}_0$, *where* $\overline{Y}_t = \frac{1}{n_t}\sum_{i:T_i=t} Y_i$ *and* $n_t = \sum_{i=1}^n \mathbf{1}\{T_i = t\}$.

(i) ***(Finite population / SATE)*** *Under a completely randomized design (CDR) with fixed* n_1 *and* n_0, *conditioning on the fixed potential outcomes* $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$,

$$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] \;=\; \tau_{\text{SATE}} \;=\; \frac{1}{n}\sum_{i=1}^n \big(Y_i(1) - Y_i(0)\big).$$

(ii) ***(Super-population / PATE)*** *If, in addition, the* n *units are drawn as an i.i.d. simple random sample from a population where* $(Y(1), Y(0))$ *are random, then*

$$\mathbb{E}[\hat{\tau}] \;=\; \tau_{\text{PATE}} \;=\; \mathbb{E}\big[Y(1) - Y(0)\big].$$

Remark 1 *We also have that* $\mathbb{E}[\hat{\sigma}_t^2] = \sigma_t^2$ *by the usual unbiasdness of the sample variance.*

Proof: (i) *Finite population.* Using linearity of expectation and the fact that the potential outcomes are fixed given $\mathcal{O}_n$,

$$\mathbb{E}\big[\overline{Y}_1 \mid \mathcal{O}_n\big] = \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n T_i\, Y_i(1) \,\Big|\, \mathcal{O}_n\right] = \frac{1}{n_1}\sum_{i=1}^n Y_i(1)\mathbb{E}[T_i \;\mid\; \mathcal{O}_n] = \frac{1}{n_1}\sum_{i=1}^n Y_i(1)\frac{n_1}{n} = \frac{1}{n}\sum_{i=1}^n Y_i(1).$$

The equality follows because, by symmetry of the CRD with fixed n_1, each i is selected into treatment with probability n_1/n,

$$\mathbb{E}[T_i \;\mid\; \mathcal{O}_n] = \mathbb{E}[T_i] = \Pr(T_1 = 1) = \frac{n_1}{n}$$

Analogously,

$$\mathbb{E}\big[\overline{Y}_0 \mid \mathcal{O}_n\big] = \frac{1}{n}\sum_{i=1}^n Y_i(0).$$

Subtracting yields

$$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n}\sum_{i=1}^n \big(Y_i(1) - Y_i(0)\big) = \tau_{\text{SATE}}.$$

(ii) *Super-population.* Using the law of iterated expectation and (i),

$$\mathbb{E}[\hat{\tau}] = \mathbb{E}[\;\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n]\;] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \big(Y_i(1) - Y_i(0)\big)\right] = \mathbb{E}\big[Y(1) - Y(0)\big] = \tau_{\text{PATE}},$$

where the last equality uses i.i.d. sampling so that the expectation of the sample mean equals the population mean. $\square$

Lemma 2 (Variance of the difference-in-means estimator) *Let* $\hat{\tau} = \overline{Y}_1 - \overline{Y}_0$, *where* $\overline{Y}_t = \frac{1}{n_t}\sum_{i:T_i=t} Y_i$ *and* $n_t = \sum_{i=1}^n \mathbf{1}\{T_i = t\}$, *under a completely randomized design (CRD) with fixed* n_1 *and* n_0.

(i) ***(Finite population / SATE)*** *Conditioning on* $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$,

$$\operatorname{Var}(\hat{\tau} \mid \mathcal{O}_n) \;=\; \frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n} \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0},$$

with $S_*^2 = S_1^2 + S_0^2 - 2S_{01} \geq 0$ *and* $\mathbb{E}\left[S_*^2\right] = \operatorname{Var}\big(Y_i(1) - Y_i(0)\big)$.

(ii) ***(Super-population / PATE)*** *If, in addition, the* n *units are drawn i.i.d. from a population where* $(Y(1), Y(0))$ *are random,*

$$\operatorname{Var}(\hat{\tau}) \;=\; \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}\,.$$

Remark 2 *The above lemma illustrates why we can use the estimator for the super-population variance for the finite-population variance. In expectation this estimator yields.*

$$\mathbb{E}[\widehat{\mathrm{Var}}(\hat\tau) \mid \mathcal{O}_n] = \mathbb{E}\left[\frac{\hat\sigma_1^2}{n_1} + \frac{\hat\sigma_0^2}{n_0}\right] = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} \geq \mathrm{Var}(\hat\tau \mid \mathcal{O}_n)$$

Proof: (i) *Finite population.* The first equality follows from the review questions. We show the second equality here. By definition $2S_{01} = S_1^2 + S_0^2 - S_*^2$. Inserting into the variance

$$\begin{aligned}
\mathrm{Var}(\hat\tau \mid \mathcal{O}_n) &= \frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right) \\
&= \frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + S_1^2 + S_0^2 - S_*^2\right) \\
&= \frac{1}{n}\left(\frac{n_0+n_1}{n_1}S_1^2 + \frac{n_1+n_0}{n_0}S_0^2 - S_*^2\right) \\
&= \frac{1}{n}\left(\frac{n}{n_1}S_1^2 + \frac{n}{n_0}S_0^2 - S_*^2\right) \\
&= \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n}.
\end{aligned}$$

This proves the second equality. It remains to show that $S_*^2 \geq 0$. By Cauchy-Schwarz we have

$$S_{01} \leq S_1 S_0.$$

Noting that $S_t \geq 0$ we have

$$S_*^2 = S_1^2 + S_0^2 - 2S_{01} \geq S_1^2 + S_0^2 - 2S_1S_0 = (S_1 - S_0)^2 \geq 0,$$

which proves the last claim.

(ii) *Super-population.* By the law of iterated iterated variance and our i.i.d. assumption:

$$\begin{aligned}
\mathrm{Var}(\hat\tau) &= \mathbb{E}\big[\mathrm{Var}(\hat\tau \mid \mathcal{O}_n)\big] + \mathrm{Var}\big(\mathbb{E}[\hat\tau \mid \mathcal{O}_n]\big) \\
&= \mathbb{E}\left[\frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\star^2}{n}\right] + \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n \big(Y_i(1) - Y_i(0)\big)\right) \\
&= \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0} - \frac{1}{n}E[S_\star^2] + \frac{1}{n}\mathrm{Var}(Y_i(1) - Y_i(0))
\end{aligned}$$

since $\mathbb{E}[S_t^2] = \sigma_t^2$ by the unbiasedness of the sample variance. It remains to show that $E[S_\star^2] = \mathrm{Var}(Y(1) - Y(0))$. Let $\mu_1 = \mathbb{E}[Y(1)]$, $\mu_0 = \mathbb{E}[Y(0)]$. Expand

$$S_{01} = \frac{1}{n-1}\sum_{i=1}^n \big(Y_i(1) - \overline{Y(1)}\big)\big(Y_i(0) - \overline{Y(0)}\big) = \frac{1}{n-1}\left[\sum_{i=1}^n \big(Y_i(1) - \mu_1\big)\big(Y_i(0) - \mu_0\big) - n\big(\overline{Y(1)} - \mu_1\big)\big(\overline{Y(0)} - \mu_0\big)\right]$$

Taking expectations,

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^n \big(Y_i(1) - \mu_1\big)\big(Y_i(0) - \mu_0\big)\right] &= n\mathrm{Cov}(Y_i(1), Y_i(0)) \\
\mathbb{E}\Big[n\big(\overline{Y(1)} - \mu_1\big)\big(\overline{Y(0)} - \mu_0\big)\Big] &= n\,\mathrm{Cov}\big(\overline{Y(1)}, \overline{Y(0)}\big) \\
&= n\frac{1}{n^2}\sum_{i,j=1}^n \mathrm{Cov}(Y_i(1), Y_j(0)) \\
&= \frac{1}{n}\sum_{i=1}^n \mathrm{Cov}(Y_i(1), Y_i(0)) \\
&= \mathrm{Cov}(Y_i(1), Y_i(0))
\end{aligned}$$

Therefore

$$\mathbb{E}[S_{01}] = \frac{1}{n-1}(n-1)\mathrm{Cov}(Y_i(1), Y_i(0)) = \mathrm{Cov}(Y_i(1), Y_i(0))$$

By linearity of expectation,

$$\mathbb{E}[S_*^2] = \mathbb{E}[S_1^2] + \mathbb{E}[S_0^2] - 2\,\mathbb{E}[S_{01}] = \sigma_1^2 + \sigma_0^2 - 2\mathrm{Cov}(Y_i(1), Y_i(0)).$$

Finally, for any two random variables A, B, $\mathrm{Var}(A - B) = \mathrm{Var}(A) + \mathrm{Var}(B) - 2\mathrm{Cov}(A, B)$. With $A = Y(1)$ and $B = Y(0)$,

$$\mathrm{Var}\big(Y(1) - Y(0)\big) = \sigma_1^2 + \sigma_0^2 - 2\mathrm{Cov}(Y_i(1), Y_i(0)).$$

Thus $\mathbb{E}[S_*^2] = \mathrm{Var}\big(Y(1) - Y(0)\big)$, as claimed.

$\square$

	Finite-Population	**Super-Population**
Population	Set of units in the study is fixed and finite	Units are drawn i.i.d. from a large or infinite "super-population."
Potential Outcomes	Fixed set $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$	$(Y_i(0), Y_i(1))$ i.i.d. $\sim \mathbb{P}^*$
Inference	Treatment effects for this specific group of units	Treatment effects for the underlying population distribution
Source of Randomness	Treatment assignment	Treatment assignment Sampling of units from the population
Random Variables	Treatment T_i	Treatment and Potential Outcomes $T_i, Y_i(0), Y_i(1)$
Example	What is the ATE on these n units in our experiment?	What is the ATE in the population from which these n units were sampled?
Validity	Internal	External
Estimand	$\tau_{\text{SATE}} = \frac{1}{n}\sum_{i=1}^n \big(Y_i(1) - Y_i(0)\big)$	$\tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)]$
Estimator	Difference in means $\hat{\tau} = \overline{Y}_1 - \overline{Y}_0$	
Unbiasedness	$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{\text{SATE}}$	$\mathbb{E}[\hat{\tau}] = \tau_{\text{PATE}}$
Variance of Estimator	$\text{Var}(\hat{\tau} \mid \mathcal{O}_n) =$ $\frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right)$	$\text{Var}(\hat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}$
Estimator of Variance	$\widehat{\text{Var}}(\hat{\tau}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}$ Finite-Population: Conservative estimator $\mathbb{E}[\widehat{\text{Var}}(\hat{\tau}) \mid \mathcal{O}_n] \geq \text{Var}(\hat{\tau} \mid \mathcal{O}_n)$, equal to design variance iff $Y_i(1) - Y_i(0) = c$ for all i, Super-Population: Unbiased estimator $\mathbb{E}[\widehat{\text{Var}}(\hat{\tau})] = \text{Var}(\hat{\tau})$	
Asymptotics $\frac{n_1}{n} \to p \in (0,1)$	Finite-population CLT: $\dfrac{\hat{\tau} - \tau_{\text{SATE}}}{\sqrt{\frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right)}} \xrightarrow{d} \mathcal{N}(0,1)$	Super-population LLN and CLT: Consistency (LLN): $\hat{\tau} \xrightarrow{p} \tau_{\text{PATE}}$ CLT: $\dfrac{\hat{\tau} - \tau_{\text{PATE}}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}}} \xrightarrow{d} \mathcal{N}(0,1)$
CI in practice	s.e. $= \sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}}$ $(1-\alpha) \times 100\%$ CI: $\left[\hat{\tau} \pm z_{1-\alpha/2} \cdot \text{s.e.}\right]$ Conservative in finite-population; Unbiased in super-population	

Assumptions

The Table is valid under SUTVA and a completely randomized design (CRD).

Notation

$$n_1 = \sum_{i=1}^n T_i, \quad n_0 = n - n_1$$

$$\overline{Y(t)} = \frac{1}{n}\sum_{i=1}^n Y_i(t), \qquad \overline{Y}_t = \frac{1}{n_t}\sum_{i:T_i=t} Y_i,$$

$$S_t^2 = \frac{1}{n-1}\sum_{i=1}^n \left(Y_i(t) - \overline{Y(t)}\right)^2, \quad S_{01} = \frac{1}{n-1}\sum_{i=1}^n \big(Y_i(1) - \overline{Y(1)}\big)\big(Y_i(0) - \overline{Y(0)}\big),$$

$$\sigma_t^2 = \text{Var}\big(Y_i(t)\big) = \mathbb{E}[S_t^2], \qquad \hat{\sigma}_t^2 = \frac{1}{n_t - 1}\sum_{i:T_i=t}\left(Y_i - \overline{Y}_t\right)^2.$$

Finite-Population Bounds for the Variance using Cauchy Schwarz

Since S_{01} is unobserved, we can obtain these (tight without further assumptions) bounds on the variance for the τ_{SATE}

$$\frac{n_0 n_1}{n}\left(\frac{S_1}{n_1} - \frac{S_0}{n_0}\right)^2 \leq \text{Var}(\hat{\tau} \mid \mathcal{O}_n) \leq \frac{n_0 n_1}{n}\left(\frac{S_1}{n_1} + \frac{S_0}{n_0}\right)^2$$